

2026-1 Computer Graphics
Programming Assignment 3

1 Display multiple 3D polygonal models with textures.

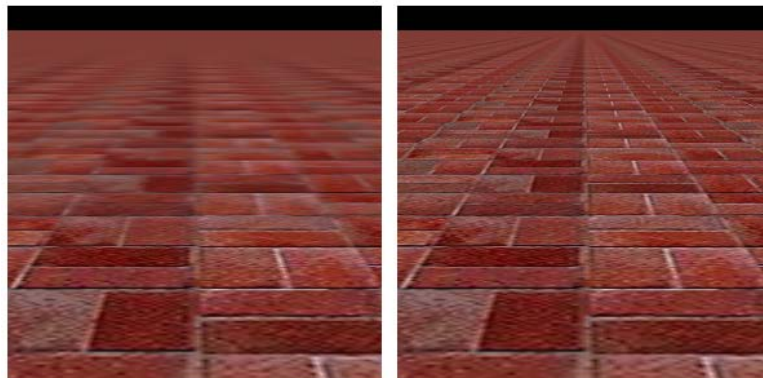
- 1.1 Load **three** different **3D** models from **OBJ** files. [2 pt]
 - ✓ Do **not** use simple models (primitives) such as cube, cylinder, pyramid, sphere, etc.
 - ✓ All models should be displayed without any **hole**.
- 1.2 Apply a different **texture** to each model. [2 pt]
 - ✓ Do **not** use a single color (e.g., red, blue, green, etc.) for texture.
- 1.3 **Self-rotate** (자전) each model **continuously** around a **y-axis** at different **positions**. [1 pt]

2 Display a ground plane with a texture.

- 2.1 Create a **textured plane** model by loading from an OBJ file or specifying vertices. [1 pt]
 - ✓ Apply a **patterned** image for the texture.
- 2.2 Set up the **camera** such that the plane represents a **ground with horizon**. [1 pt]

3 Display filtering effects using keyboard inputs.

- 3.1 **3/4** keys: apply the linear/anisotropic filter mode. [1 pt]
- 3.2 The visual differences should be **clearly** displayed between two modes. [2 pt]
 - ✓ Show **blurred** patterns in the linear mode and **clarified** patterns in the anisotropic mode.



(Left: linear, Right: anisotropic)

✓ **Submission & Cautions**

- Delete all folders (**Debug, Release, ipch**) and files (***.db, *.exe**) that are **not** necessary for build.
(*폴더 보기 → 표시 → 숨김 항목 → .vs 폴더 → ipch 폴더 및 *.db 파일 삭제)
- Compress an entire project and a report (**ZIP**, size limit: **100MB**) and submit to **ClassRoom** (***not** accept via E-mail).
- Use only **Visual Studio 2022** and **DirectX 11**.
- Build the program in a **x86 (32-bit)** and **Release** mode.
- Cases of a score of **0**: not using class framework, not building source code, not executing the program, copying source code from others or a previous work, not demonstrating, submissions after the deadline.